

Syllabus

Spring semester 2026

Course Title	General Chemistry I	Course Code			Credit	4	Classroom	Seminar G
		Type			Hours	4		
Teacher	Name	Class no.	Office	Office Hour*	Contact	E-mail		
	Hahn, Sangjoon	4	Tamgu 2203	Mon. 6, 7 Tue. 5	+82-51-606-2210	eipiplus1@ksa.kaist.ac.kr		

*Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

1. Course Description

Chemistry is a discipline that studies matter, including its structure, changes, and properties, and is one of the most important fields of pure natural science alongside physics and biology. In General Chemistry 1, students begin with the fundamentals of chemistry and go on to study areas such as chemical bonding, quantum mechanics, organic chemistry, inorganic chemistry, and spectroscopy.

2. Core Target Competencies

Subject Core Competencies						
Analytical and Creative Thinking						
The ability to analyze and create ideas or knowledge.						

3. Textbooks and References

A. Textbook: Principles of modern chemistry, 8th ed. Oxtoby

4. Class Method

A. Lecture

5. Evaluation

A. Evaluation Criteria

OO activity	OO project	Attitude	Midterm exam	Final exam	Attendance	Sum
		20%	35%	35%	10%	100

B. Evaluation Method

- (1) Final grades are based on written exams, attendance, and attitude.
- (2) - 2 points deducted for unauthorized absence from attendance.
 - 5 or more unexcused absences will result in a failing grade (F).
 - Unauthorized use of computers or smartphones during class will result in a 5-point deduction from attitude.

6. Criteria for Awarding Credits

 **F : In the case of receiving a score of 50 or less out of 100 according to the criteria of the evaluation method.**

Scoring for Missed Examinations (Authorized/Medical Absence):

: The acknowledged score for an exam missed due to an authorized or medical absence shall be calculated as follows: (The score at the same percentile as the student's overall percentile rank from other exams taken in the same subject) × (Authorized Absence: 100%, Medical Absence: 80%)

 **Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.**

- Attendance for at least three-quarters of the total class hours is required to earn credit.
- For Pass/Fail courses, if the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
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- Calculation of the basis score for awarding credit to absentees :
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- If a basis score cannot be calculated, the grading method :
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- ※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

Week (period)	Topic and Contents	(Notes)
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Week 1 2.23.~3.1.	Chapter 1. The Atom in modern Chemistry	
Core Competencies	♦ Analytical and Creative Thinking	
Week 2 3.2.~3.8.	Chapter 3. Chemical Bonding: The Classical Description	3.2. Substitute Holiday
Core Competencies	♦ Analytical and Creative Thinking	
Week 3 3.9.~3.15.	Chapter 4. Introduction to quantum mechanics	
Core Competencies	♦ Analytical and Creative Thinking	
Week 4 3.16.~3.22.	Chapter 4. Introduction to quantum mechanics Chapter 5. Quantum mechanics and atomic structure	
Core Competencies	♦ Analytical and Creative Thinking	
Week 5 3.23.~3.29.	Chapter 5. Quantum mechanics and atomic structure	
Core Competencies	♦ Analytical and Creative Thinking	
Week 6 3.30.~4.5.	Chapter 6. Quantum mechanics and molecular structure	
Core Competencies	♦ Analytical and Creative Thinking	
Week 7 4.6.~4.12.	Chapter 6. Quantum mechanics and molecular structure Chapter 7. Bonding in Organic Molecules	
Core Competencies	♦ Analytical and Creative Thinking	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)	4.13. Regular Classes

Week 9 4.20.~4.26.	Chapter 8. Bonding in Transition Metal Compounds and Coordination Complexes	
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Week 10 4.27.~5.3.	Chapter 9. The Gaseous State	
Core Competencies	♦ Analytical and Creative Thinking	
Week 11 5.4.~5.10.	Chapter 10. Solids, Liquids, and Phase Transitions	5.5. Children's Day
Core Competencies	♦ Analytical and Creative Thinking	
Week 12 5.11.~5.17.	Chapter 11. Solutions	
Core Competencies	♦ Analytical and Creative Thinking	
Week 13 5.18.~5.24.	Chapter 20. Molecular Spectroscopy and Photochemistry - Introduction	
Core Competencies	♦ Analytical and Creative Thinking	
Week 14 5.25.~5.31.	Chapter 20. Molecular Spectroscopy and Photochemistry - Rotational and Vibrational Spectroscopy	5.25. Substitute Holiday
Core Competencies	♦ Analytical and Creative Thinking	
Week 15 6.1.~6.7.	Chapter 20. Molecular Spectroscopy and Photochemistry - NMR and Electronic Spectroscopy	6.3. Local Election Day
Core Competencies	♦ Analytical and Creative Thinking	
Week 16 6.8.~6.14.	Make-up class week	
Week 17 6.15.~6.19.	Final Exams (6.18.~6.19.)	

※ The above plans may change according to the academic schedule.

[Reference] Syllabus Competence Reference

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/Quantitative Literacy	5. Global Perspective	6. Adaptability/Progression / Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/Mindset
Ability to identify, manage and deal with complex problems	Ability to understand and express themselves in two or more languages	Ability to be the first to come up with and present new ideas	Ability to understand, use and apply digital technology	Ability to develop an open mind about the traditions or values of others	Ability to develop flexibility, agility and adaptability	Ability to sustain an empathetic and understanding perspective	Sincerity
Ability to see through prejudice and identify reliable information and uncertain information	Ability to communicate clearly with diverse audiences or readers	Ability to influence people to follow themselves, rather than lead them through directives, orders, or authority	Ability to create digital contents and media	Ability to understand the history, politics, religion, and culture of another region	Ability to challenge unfamiliar situations	Ability to cultivate sincerity, honesty, fairness, and respect creativity	Creativity
Ability to find quality information in the flood of information	Ability to listen attentively	Ability to build trust, resolve conflicts, and support others	Ability to utilize multimedia resources to effectively communicate one's ideas in a variety of formats	Ability to use one or more of the international languages used worldwide	Ability to explore and experiment	Ability to display moral courage in the face of injustice	A love of learning and curiosity
Ability to articulate meaningful questions	Ability to speak effectively and coherently	Ability to facilitate group discussions or discussions, build consensus, and negotiate outcomes	Ability to master and use higher mathematics	Ability to use technology to connect people and things around the world	Ability to word effectively in situations of ambiguous or changing priorities	Ability to act responsibly for the benefit and happiness of the greater community	Resilience/elasticity
Ability to analyze and create ideas or knowledge	Ability to write concisely and clearly, considering the diversity of readers	Ability to teach, guide, and counsel Ability to seek help from others	Ability to understand traditional or recently emerging fields such as mathematics, science, technology, environmental science, robotics, geometry, cellular automata, nanotechnology, and biotechnology	Social and intellectual skills to effectively navigate information across multiple cultures	Ability to see failure as a lesson and to accept that innovation is made up of frequent failures and small successes	Ability to have an intrinsic understanding of ethical issues or dilemmas about new media or new technologies	Perseverance/patience
Ability to devise and test solutions to problems by	Ability to explain information and persuasively	Ability to collaborate on tasks, manage groups, and		Ability to use 21 st century core competencies such as communication	Ability to develop a sense of independence to seek new roles,	Ability to make rational and ethical decisions on complex	Self-efficacy

서식 있음: 왼쪽, 단어 잘림 허용

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/Quantitative Literacy	5. Global Perspective	6. Adaptability/Progression / Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/Mindset
trial and error	engage others in an engaging way	delegate responsibilities		skills, collaboration skills, critical thinking skills, creativity to properly understand and deal with global issues	ideas, or strategies	issues	
Ability to think of different ways to cope		Ability to listen and practice goals		Ability to learn from and collaborate with people from different cultures, religions and lifestyle through mutual respect and open dialogue	Ability to develop the knowledge and competencies necessary to start a business and run a business		Stress management skills
Ability to develop knowledge and perspectives that are not limited to one field but span multiple fields		Ability to share credit for success		Ability to generate diverse ideas and exploit social and cultural differences to get things done			Time management skills
Ability to make sustained inferences							
Ability to put things together or adapt them to new uses and situations							
Ability to solve new problems that cannot be solved through established rules							
Ability to use knowledge or creativity to solve complex real-world problems							

서식 있음: 왼쪽, 단어 잘림 허용

1. Complex Problem-Solving Skills	Ability to solve new and unclear problems that may arise in complex real-world situations
2. Critical Thinking Skills	Ability to compare and analyze the strengths and weaknesses of various solutions or approaches to a given problem using logic and reasoning
3. Creativity	Ability to come up with novel ideas when given a certain topic or situation or to suggest a creative method in the process of solving a problem
4. Human Resource Management Skills	Ability to select the most suitable person for a specific task and manage them to achieve maximum work performance through motivation, <u>and</u> systematic and efficient management
5. Ability to Collaborate	Ability to adjust one's own behavior in response to the behavior of others
6. Emotional Ability	Ability to perceive other people's reactions and understand why they react the way they do
7. Judgement and Decision-making Skills	Ability to make the most rational judgement and decision by analyzing the costs and benefits of a specific situation or problem
8. Orientated Towards Service	Ability to actively seek ways to help others
9. Negotiation Skills	Ability to bring together parties who differ in ideas or interests and try to reduce those differences
10. Cognitive Flexibility	Ability to apply different kinds of rules or principles to rearrange and reorganize things in different ways